Instructor Performance and Course Quality Evaluation on EduPro

Objective

This project evaluates instructor effectiveness, course quality, and enrollment behavior on the EduPro platform using exploratory data analysis and visualization techniques.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

Dataset Sheets

- Teachers
- Courses
- Transactions
- Users

Import Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
plt.style.use('ggplot')
```

Load Dataset

In [2]:
```
excel_file = pd.ExcelFile("/content/EduPro Dataset.xlsx")
print(excel_file.sheet_names)
```

```
['Users', 'Teachers', 'Courses', 'Transactions']
```

In [3]:
```
teachers = pd.read_excel("/content/EduPro Dataset.xlsx", sheet_name="Teachers")
courses = pd.read_excel("/content/EduPro Dataset.xlsx", sheet_name="Courses")
transactions = pd.read_excel("/content/EduPro Dataset.xlsx", sheet_name="Transactio
users = pd.read_excel("/content/EduPro Dataset.xlsx", sheet_name="Users")
```

Dataset Overview

In [4]:
```
print("Teachers Shape:", teachers.shape)
print("Courses Shape:", courses.shape)
```

```
print("Transactions Shape:", transactions.shape)
print("Users Shape:", users.shape)
```

```
Teachers Shape: (60, 7)
Courses Shape: (60, 8)
Transactions Shape: (10000, 7)
Users Shape: (3000, 5)
```

In [5]:
```
teachers.head()
```

Out[5]:

	TeacherID	TeacherName	Age	Gender	Expertise	YearsOfExperience	TeacherRating
0	TC00001	Leonard Montgomery	44	Female	Cybersecurity	6	3.24
1	TC00002	Jill Day	32	Female	Digital Marketing	9	4.14
2	TC00003	Amber Torres	32	Male	Design	4	1.56
3	TC00004	Kristi Scott	34	Female	Machine Learning	9	4.39
4	TC00005	David Williams	34	Male	Finance	2	3.11

Descriptive Statistics of Teachers and Courses

In [6]:
```
teachers.describe()
```

Out[6]:

	Age	YearsOfExperience	TeacherRating
count	60.000000	60.000000	60.000000
mean	38.450000	6.283333	3.125000
std	7.703015	4.715972	0.949797
min	27.000000	1.000000	1.050000
25%	32.000000	3.000000	2.487500
50%	36.500000	6.000000	3.275000
75%	46.250000	8.000000	3.827500
max	50.000000	24.000000	4.970000

In [7]:
```
courses.describe()
```

Out[7]:

	CoursePrice	CourseDuration	CourseRating
count	60.000000	60.000000	60.000000
mean	92.986333	27.632500	3.097833
std	153.601506	16.092578	1.171232
min	0.000000	1.200000	1.130000
25%	0.000000	14.500000	2.107500
50%	0.000000	28.505000	3.065000
75%	133.615000	43.012500	4.102500
max	490.900000	49.730000	4.940000

Missing Values Check

In [8]:
```
teachers.isnull().sum()
```

Out[8]:

	0
TeacherID	0
TeacherName	0
Age	0
Gender	0
Expertise	0
YearsOfExperience	0
TeacherRating	0

dtype: int64

In [9]:
```
courses.isnull().sum()
```

Out[9]:

	0
CourseID	0
CourseName	0
CourseCategory	0
CourseType	0
CourseLevel	0
CoursePrice	0
CourseDuration	0
CourseRating	0

dtype: int64

In [10]:
```
transactions.isnull().sum()
```

Out[10]:

	0
TransactionID	0
UserID	0
CourseID	0
TransactionDate	0
Amount	0
PaymentMethod	0
TeacherID	0

dtype: int64

In [11]:
```
users.isnull().sum()
```

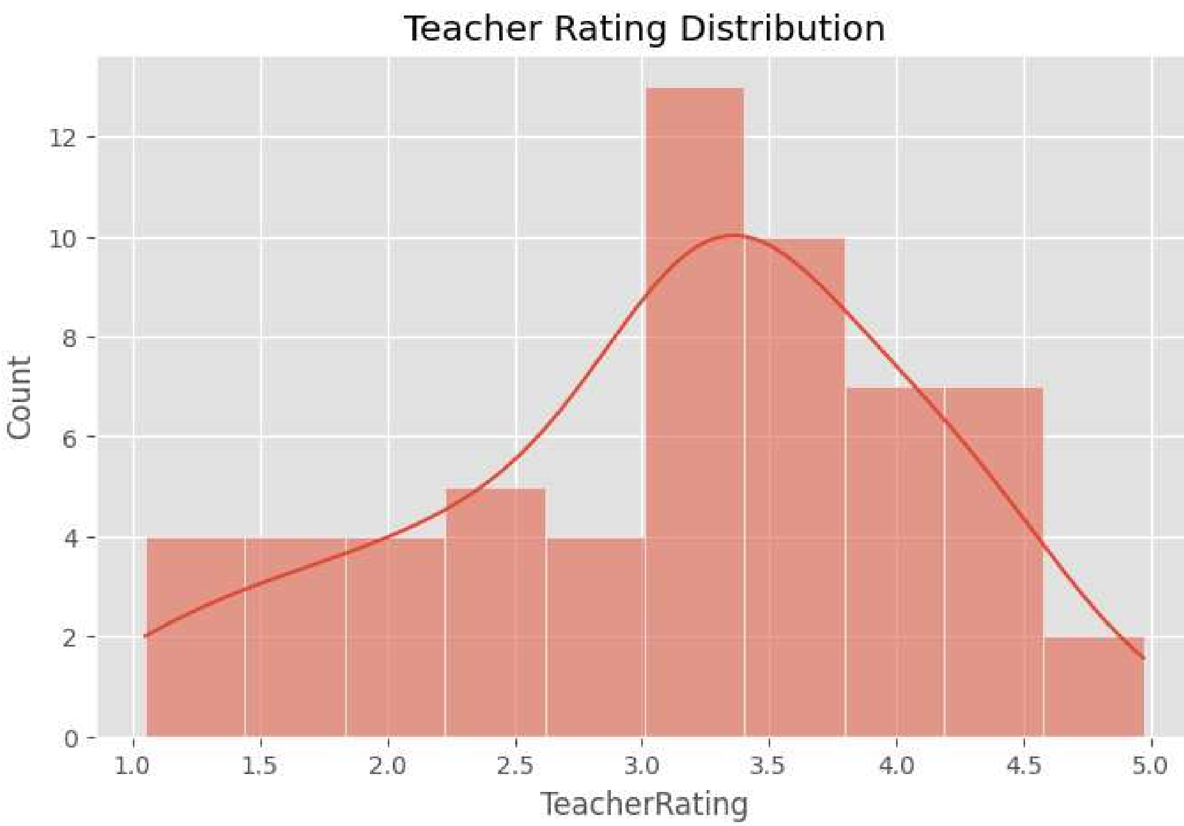
Out[11]:

	0
UserID	0
UserName	0
Age	0
Gender	0
Email	0

dtype: int64

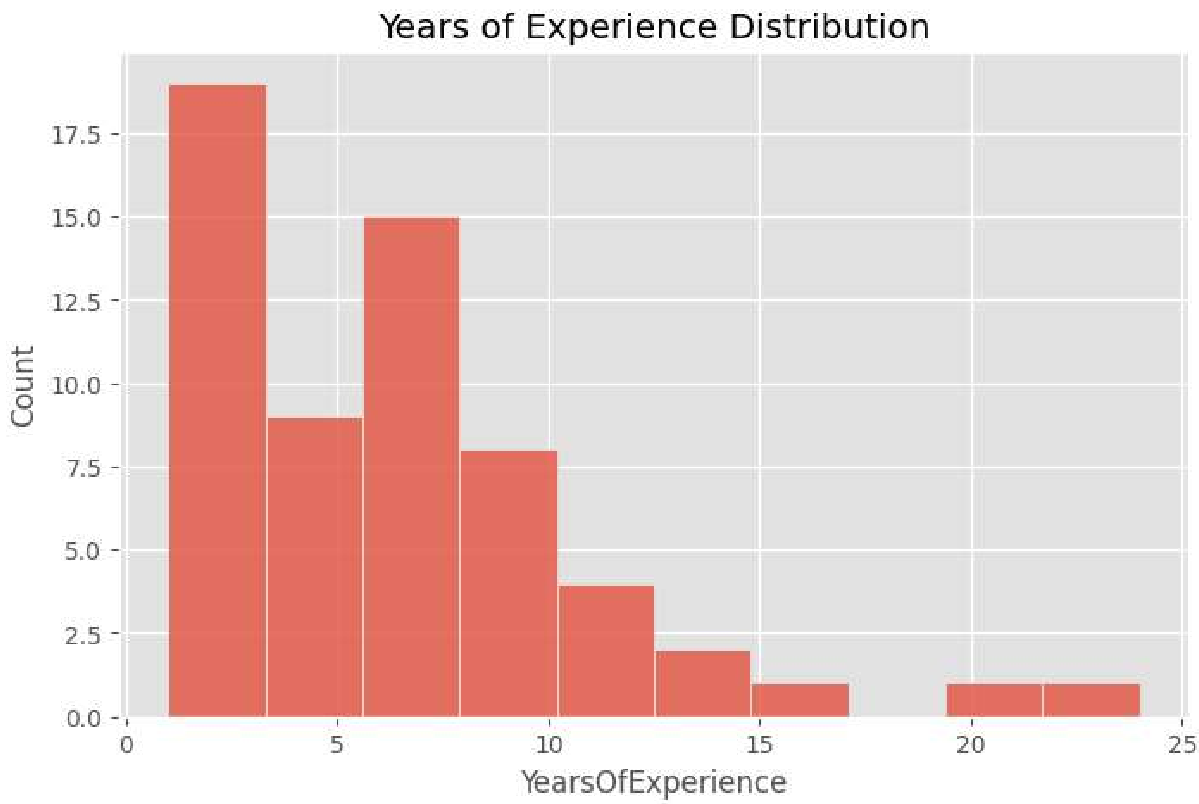
Teacher Rating Distribution

In [12]:
```
plt.figure(figsize=(8,5))
sns.histplot(
    teachers["TeacherRating"],
    bins=10,
    kde=True)
plt.title("Teacher Rating Distribution")
plt.show()
```



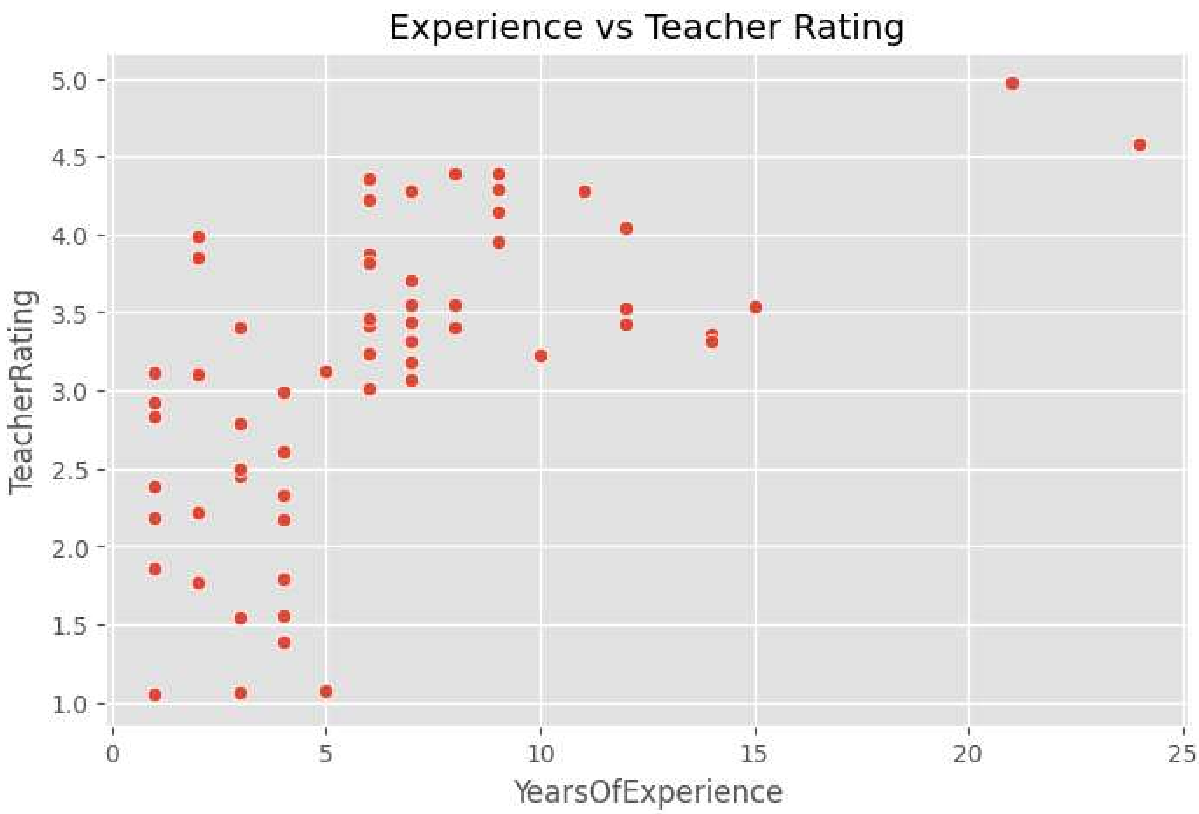
Experience Distribution

In [13]:
```
plt.figure(figsize=(8,5))
sns.histplot(
    teachers["YearsOfExperience"],
    bins=10)
plt.title("Years of Experience Distribution")
plt.show()
```



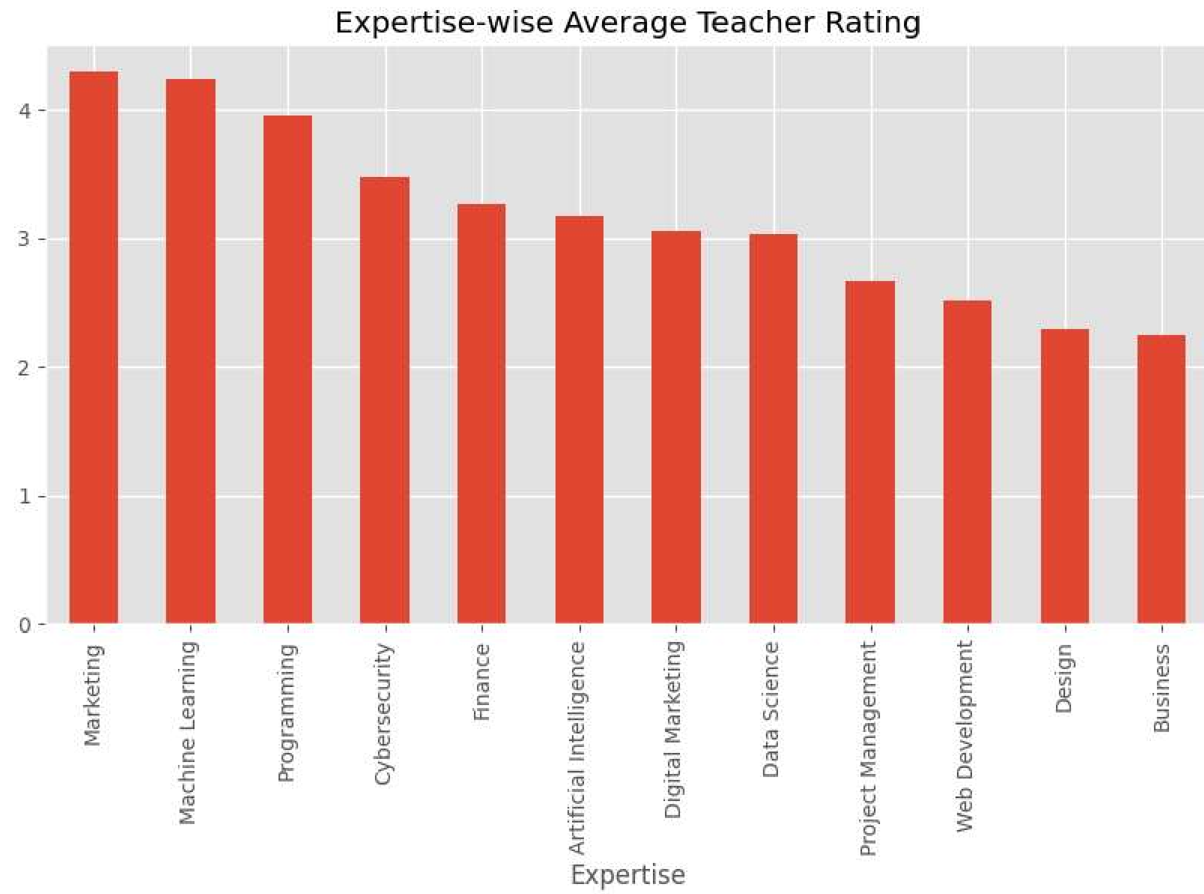
Experience vs Rating

In [14]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(
    data=teachers,
    x="YearsOfExperience",
    y="TeacherRating")
plt.title("Experience vs Teacher Rating")
plt.show()
```



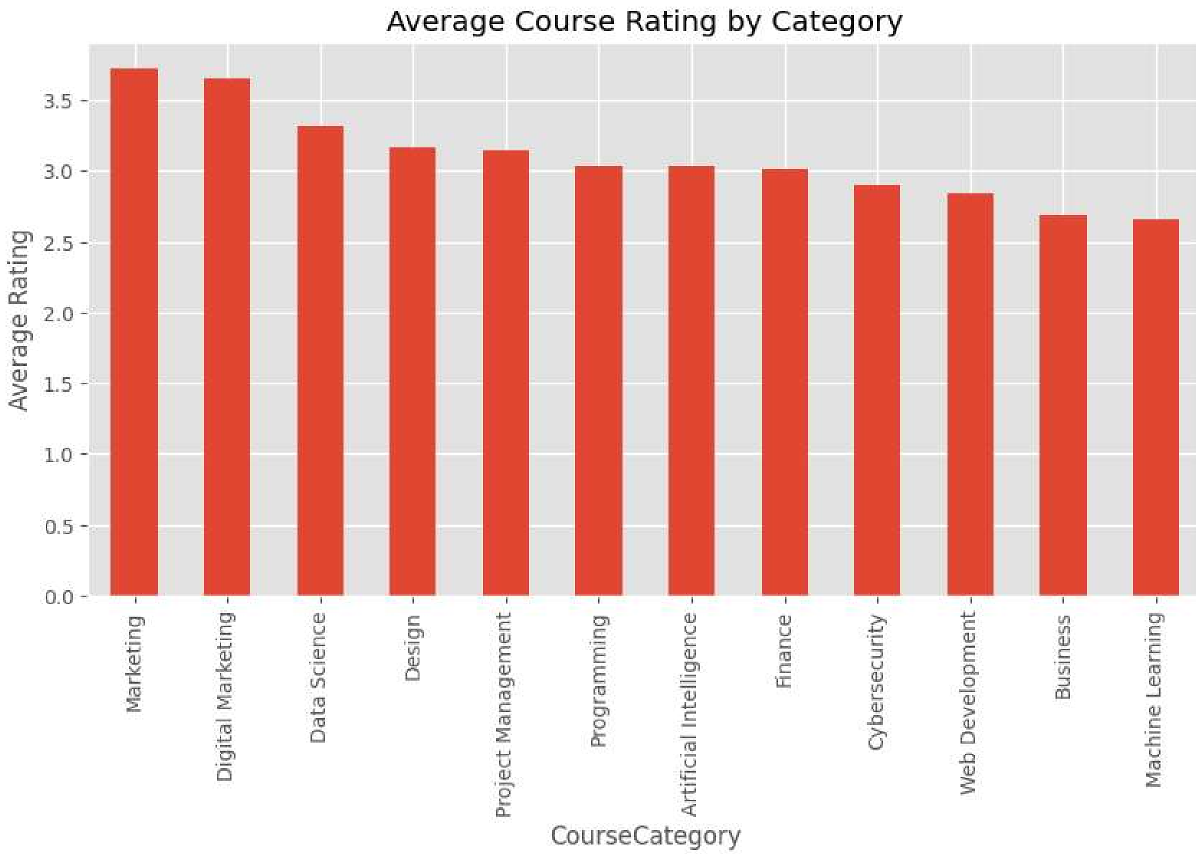
Expertise-wise Average Rating

In [15]:
```
expertise_rating = (
    teachers.groupby("Expertise")
    ["TeacherRating"]
    .mean()
    .sort_values(ascending=False))
plt.figure(figsize=(10,5))
expertise_rating.plot(kind="bar")
plt.title("Expertise-wise Average Teacher Rating")
plt.show()
```



Course Category vs Course Rating

In [16]:
```
category_rating = (
    courses.groupby("CourseCategory")
    ["CourseRating"]
    .mean()
    .sort_values(ascending=False))
plt.figure(figsize=(10,5))
category_rating.plot(kind="bar")
plt.title("Average Course Rating by Category")
plt.ylabel("Average Rating")
plt.show()
```



Course Level vs Course Rating

In [17]:

```
level_rating = (
    courses.groupby("CourseLevel")
    ["CourseRating"]
    .mean())
plt.figure(figsize=(8,5))
level_rating.plot(kind="bar")
plt.title("Average Course Rating by Level")
plt.ylabel("Average Rating")
plt.show()
```

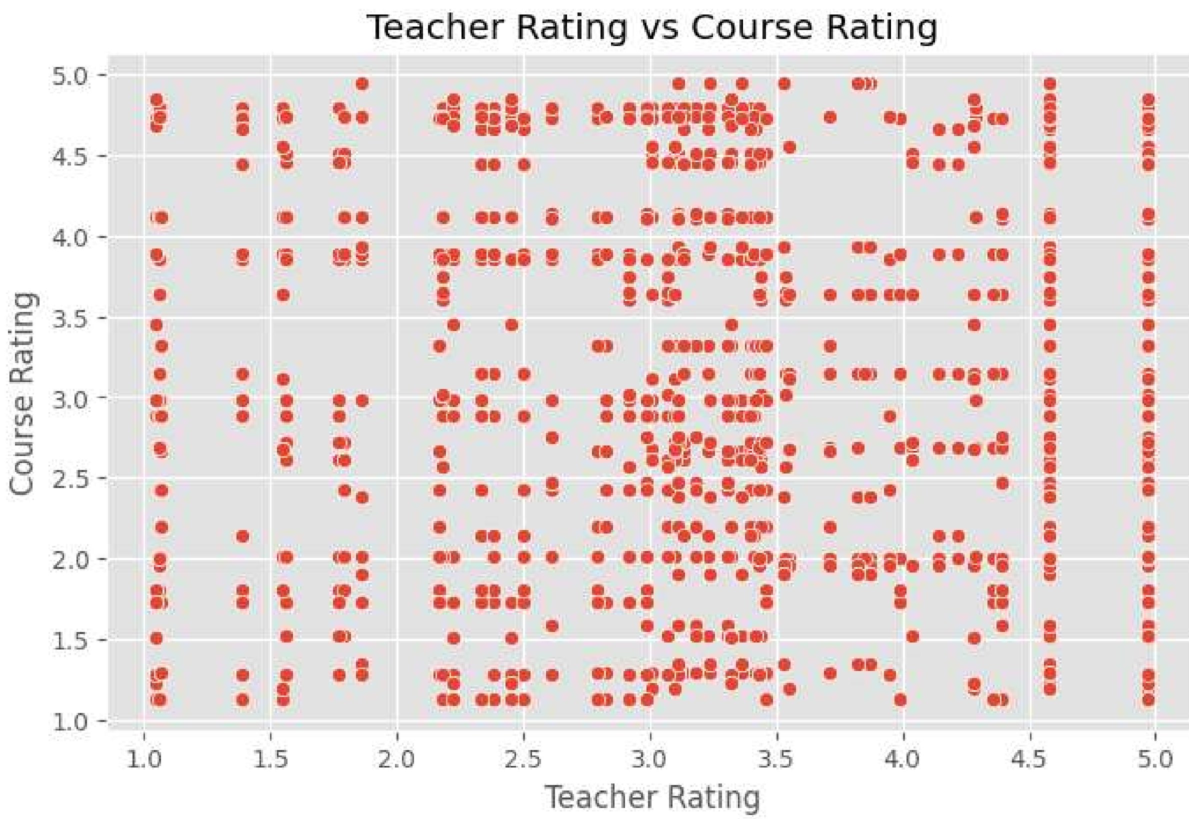
Course Quality Heatmap

In [18]:
```
heatmap_data = courses.pivot_table(
    values="CourseRating",
    index="CourseCategory",
    columns="CourseLevel",
    aggfunc="mean")
plt.figure(figsize=(10,6))
sns.heatmap(
    heatmap_data,
    annot=True,
    cmap="YlGnBu")
plt.title("Course Quality Heatmap")
plt.show()
```

Teacher Rating vs Course Rating Analysis

In [38]:

```
teacher_course = transactions.merge(
    teachers[['TeacherID', 'TeacherRating']],
    on='TeacherID').merge(
    courses[['CourseID', 'CourseRating']],
    on='CourseID')
plt.figure(figsize=(8,5))
sns.scatterplot(
    data=teacher_course,
    x='TeacherRating',
    y='CourseRating')
plt.title('Teacher Rating vs Course Rating')
plt.xlabel('Teacher Rating')
plt.ylabel('Course Rating')
plt.show()
corr_value = (
    teacher_course[['TeacherRating', 'CourseRating']]
    .corr()
    .iloc[0,1])
print(f"Pearson Correlation Coefficient: {corr_value:.3f}")
if abs(corr_value) < 0.3:
    print("Insight: Weak relationship between Teacher Rating and Course Rating")
elif abs(corr_value) < 0.7:
    print("Insight: Moderate relationship between Teacher Rating and Course Rating"
else:
    print("Insight: Strong relationship between Teacher Rating and Course Rating")
```




```
Pearson Correlation Coefficient: -0.002
Insight: Weak relationship between Teacher Rating and Course Rating
```

Analysis Insight

The correlation analysis evaluates the relationship between instructor ratings and course ratings. The results indicate that teacher ratings and course ratings exhibit a weak relationship, suggesting that course quality may also be influenced by factors such as course content, structure, learner engagement, and category-specific characteristics in addition to instructor performance.

Instructor Leaderboard

Top 10 Instructors

In [20]:
```
top_teachers = (
    teachers.sort_values(
        by="TeacherRating",
        ascending=False)
    .head(10))
top_teachers[
    ["TeacherName","Expertise","TeacherRating"]]
```

Out[20]:

	TeacherName	Expertise	TeacherRating
41	Yolanda Levine	Machine Learning	4.97
39	Kimberly Miller	Cybersecurity	4.58
3	Kristi Scott	Machine Learning	4.39
36	Debra Escobar	Finance	4.39
6	Angela Beard	Machine Learning	4.36
52	Aaron Kirby	Marketing	4.29
17	Kari Pierce	Project Management	4.28
53	Shelby Patrick	Artificial Intelligence	4.28
23	Michael Kim	Digital Marketing	4.22
1	Jill Day	Digital Marketing	4.14

Bottom 10 Teachers

In [21]:

```
bottom_teachers = (
    teachers.sort_values(
        by="TeacherRating",
        ascending=True)
    .head(10))
bottom_teachers[
    ["TeacherName","Expertise","TeacherRating"]]
```

Out[21]:

	TeacherName	Expertise	TeacherRating
42	John Williams	Project Management	1.05
27	Timothy Bailey	Business	1.06
44	Emily Gutierrez	Web Development	1.07
35	Brenda Mclean	Digital Marketing	1.39
26	Laurie Michael	Artificial Intelligence	1.55
2	Amber Torres	Design	1.56
50	John Obrien	Design	1.77
46	Brad Griffin	Design	1.79
30	Brenda Mercer	Cybersecurity	1.86
24	Justin Pham	Web Development	2.17

KPI Calculation

In [22]:
```
print("Average Teacher Rating:",
      round(teachers["TeacherRating"].mean(),2))
print("Average Course Rating:",
      round(courses["CourseRating"].mean(),2))
print("Average Experience:",
      round(teachers["YearsOfExperience"].mean(),2))
print("Total Courses:",
      len(courses))
print("Total Teachers:",
      len(teachers))
print("Total Transactions:",
      len(transactions))
```

```
Average Teacher Rating: 3.12
Average Course Rating: 3.1
Average Experience: 6.28
Total Courses: 60
Total Teachers: 60
Total Transactions: 10000
```

Descriptive Statistical Analysis

In [23]:
```
teachers.describe()
```

Out[23]:

	Age	YearsOfExperience	TeacherRating
count	60.000000	60.000000	60.000000
mean	38.450000	6.283333	3.125000
std	7.703015	4.715972	0.949797
min	27.000000	1.000000	1.050000
25%	32.000000	3.000000	2.487500
50%	36.500000	6.000000	3.275000
75%	46.250000	8.000000	3.827500
max	50.000000	24.000000	4.970000

In [24]:
```
courses.describe()
```

Out[24]:

	CoursePrice	CourseDuration	CourseRating
count	60.000000	60.000000	60.000000
mean	92.986333	27.632500	3.097833
std	153.601506	16.092578	1.171232
min	0.000000	1.200000	1.130000
25%	0.000000	14.500000	2.107500
50%	0.000000	28.505000	3.065000
75%	133.615000	43.012500	4.102500
max	490.900000	49.730000	4.940000

Additional Analysis

Merge Teachers + Transactions

In [25]:
```
teacher_transactions = transactions.merge(
    teachers,
    on="TeacherID",
    how="left")
```

Enrollment Count by Teacher

In [26]:
```
teacher_enrollment = (
    teacher_transactions.groupby(
        ["TeacherName","TeacherRating"])
    .size()
    .reset_index(name="EnrollmentCount"))
teacher_enrollment.head()
```

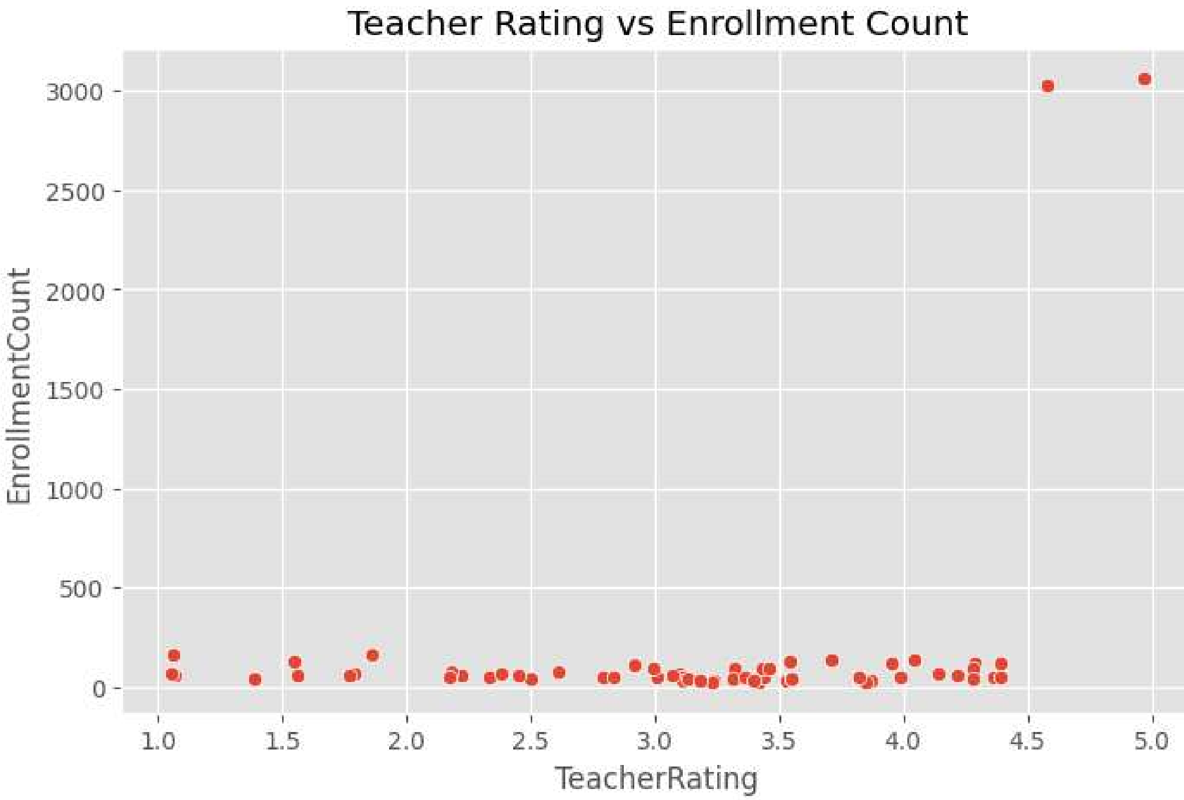
Out[26]:

	TeacherName	TeacherRating	EnrollmentCount
0	Aaron Kirby	4.29	120
1	Adam Holmes	3.99	51
2	Alexander Tucker	2.22	58
3	Alexander Williams	3.41	24
4	Alexis Everett	3.95	123

Teacher Rating vs Enrollment

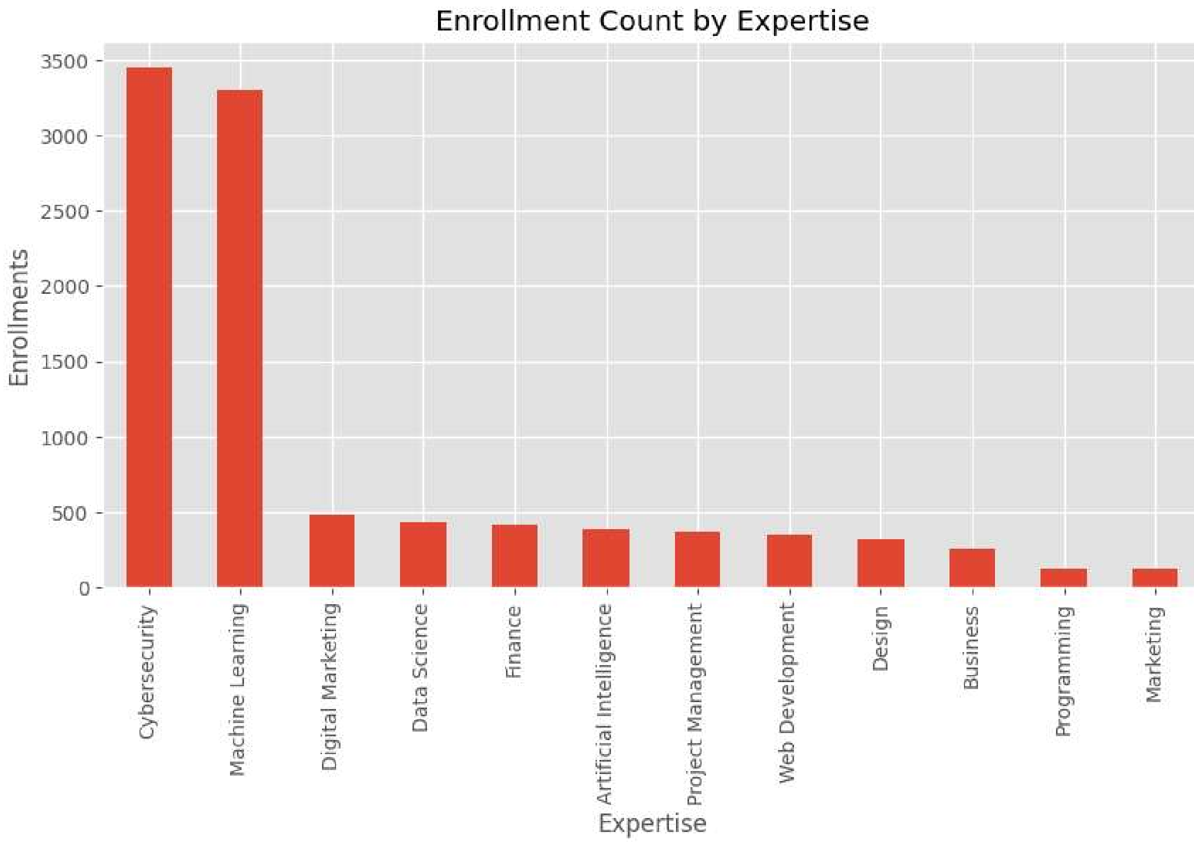
In [27]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(
    data=teacher_enrollment,
    x="TeacherRating",
```

```
    y="EnrollmentCount")
plt.title("Teacher Rating vs Enrollment Count")
plt.show()
```



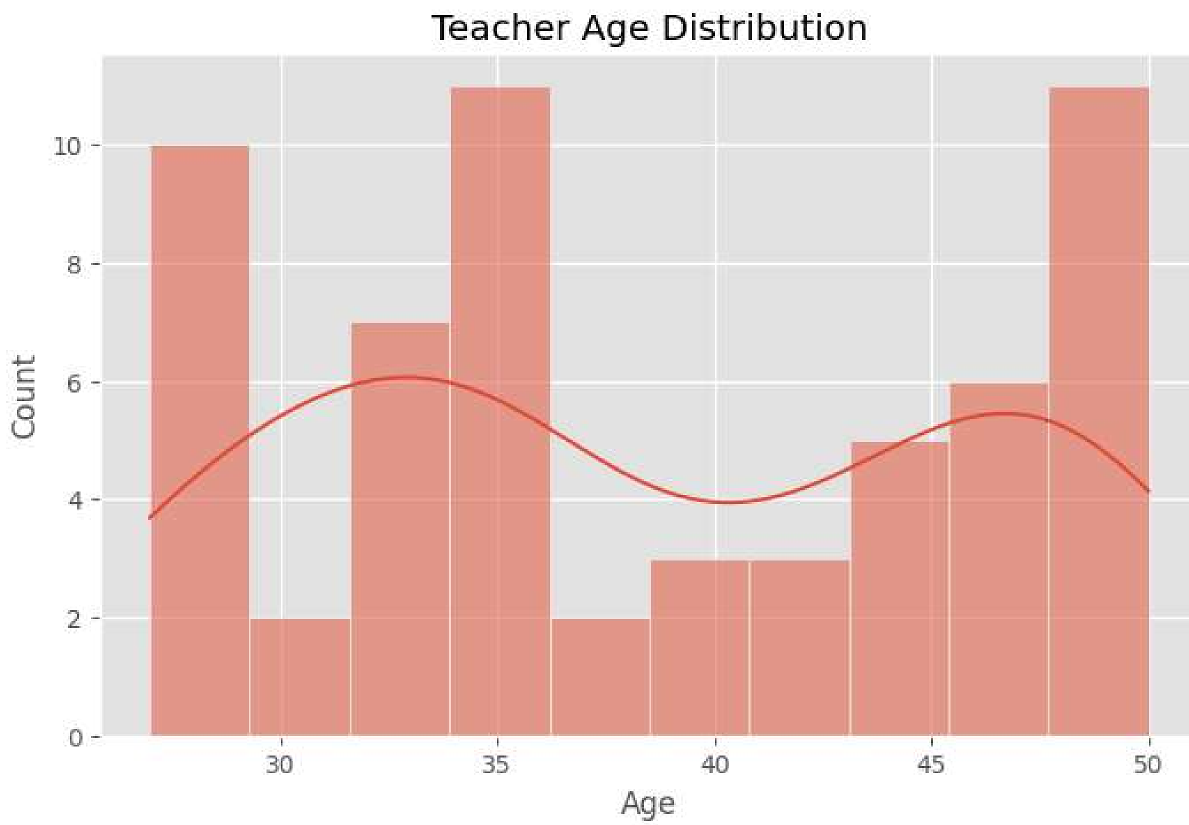
Enrollment by Expertise

In [28]:
```
expertise_enrollment = (
    teacher_transactions.groupby("Expertise")
    .size()
    .sort_values(ascending=False))
plt.figure(figsize=(10,5))
expertise_enrollment.plot(kind="bar")
plt.title("Enrollment Count by Expertise")
plt.ylabel("Enrollments")
plt.show()
```



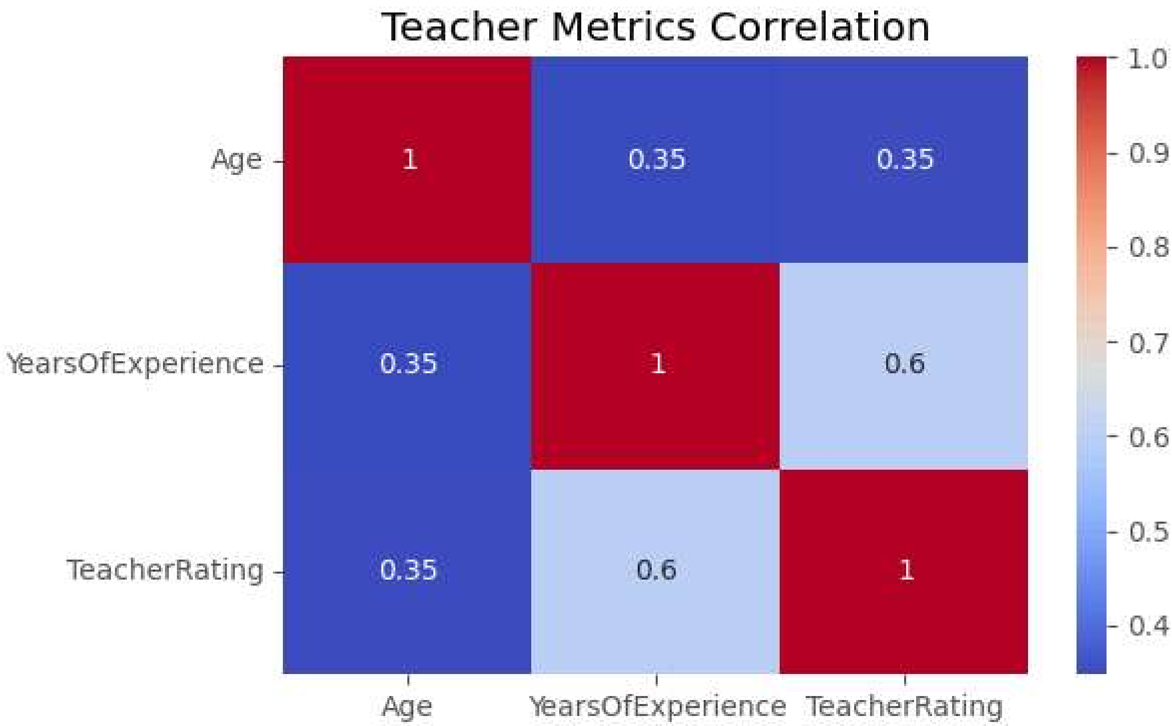
Teacher Age Distribution Analysis

In [29]:
```
plt.figure(figsize=(8,5))
sns.histplot(
    teachers["Age"],
    bins=10,
    kde=True)
plt.title("Teacher Age Distribution")
plt.show()
```



Correlation Analysis of Teacher Metrics

In [30]:
```
numeric_cols = teachers[
    ["Age",
     "YearsOfExperience",
     "TeacherRating"]]
plt.figure(figsize=(6,4))
sns.heatmap(
    numeric_cols.corr(),
    annot=True,
    cmap="coolwarm")
plt.title("Teacher Metrics Correlation")
plt.show()
```

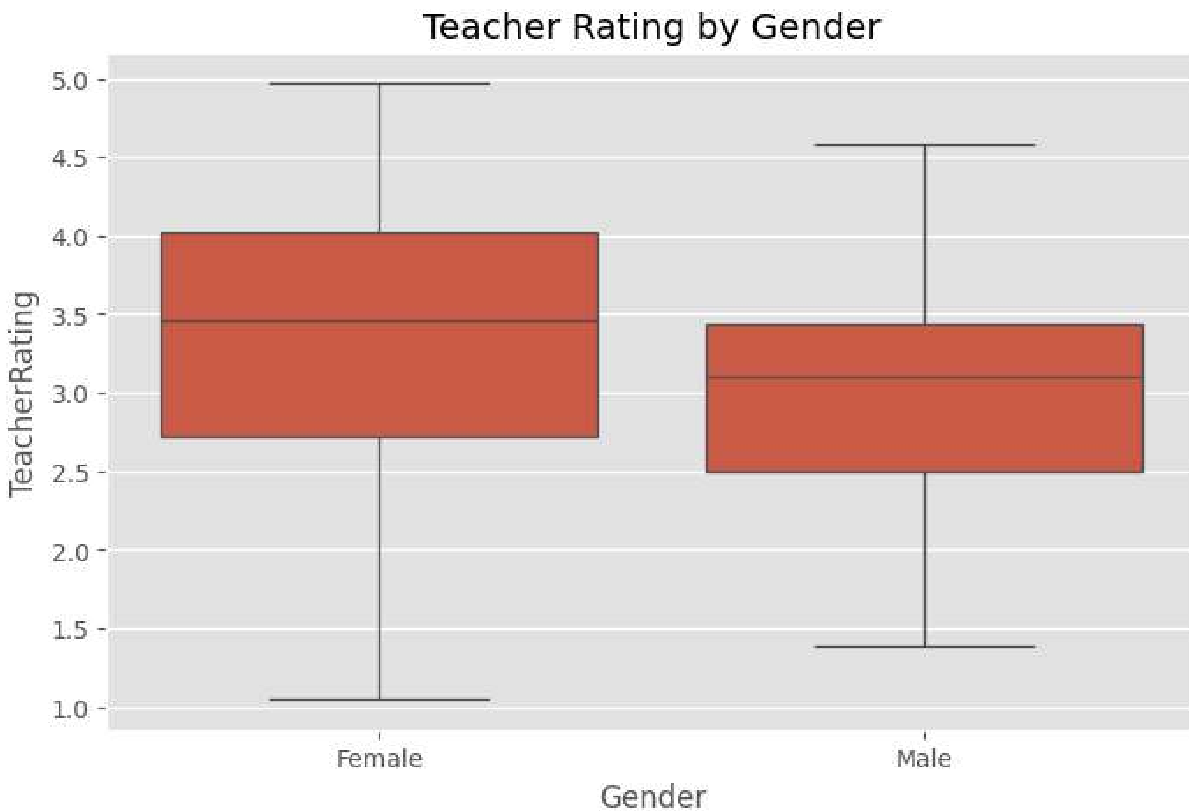


Correlation Insight

- TeacherRating shows a positive relationship with YearsOfExperience.
- Age and experience are moderately associated.
- Experience appears to contribute positively toward instructor quality.

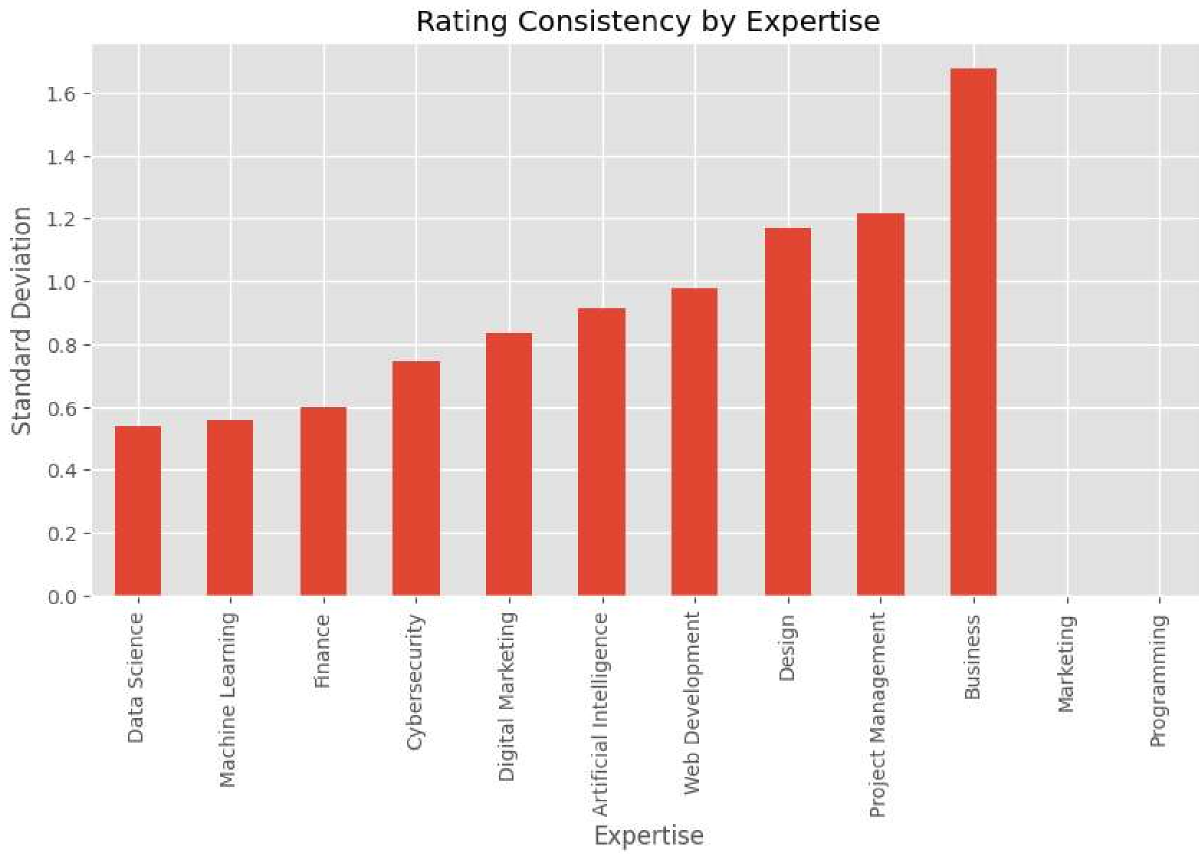
Teacher Rating Analysis by Gender

In [31]:
```
plt.figure(figsize=(8,5))
sns.boxplot(
    data=teachers,
    x="Gender",
    y="TeacherRating")
plt.title("Teacher Rating by Gender")
plt.show()
```



Rating Consistency Analysis

In [32]:
```
rating_consistency = (
    teachers.groupby("Expertise")["TeacherRating"]
    .std()
    .sort_values())
plt.figure(figsize=(10,5))
rating_consistency.plot(kind="bar")
plt.title("Rating Consistency by Expertise")
plt.ylabel("Standard Deviation")
plt.show()
rating_consistency.head()
```



Out[32]:

Expertise	TeacherRating
Data Science	0.539537
Machine Learning	0.556893
Finance	0.600591
Cybersecurity	0.745762
Digital Marketing	0.837269

dtype: float64

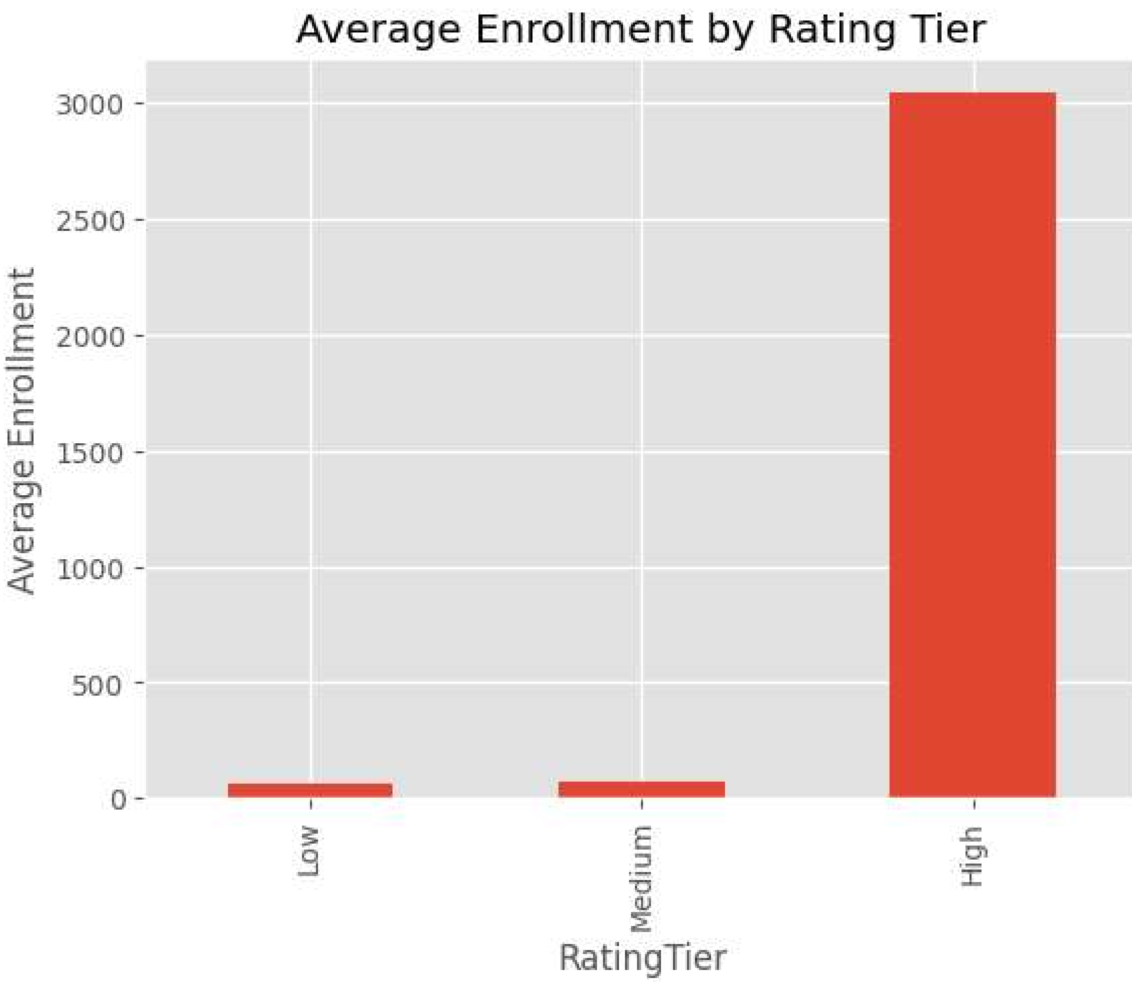
Rating Consistency Insight

Lower standard deviation indicates more consistent instructor ratings within an expertise domain. Data Science and Machine Learning exhibit the most consistent performance, while Finance demonstrates greater variation in instructor ratings.

Enrollment Analysis by Teacher Rating Tier

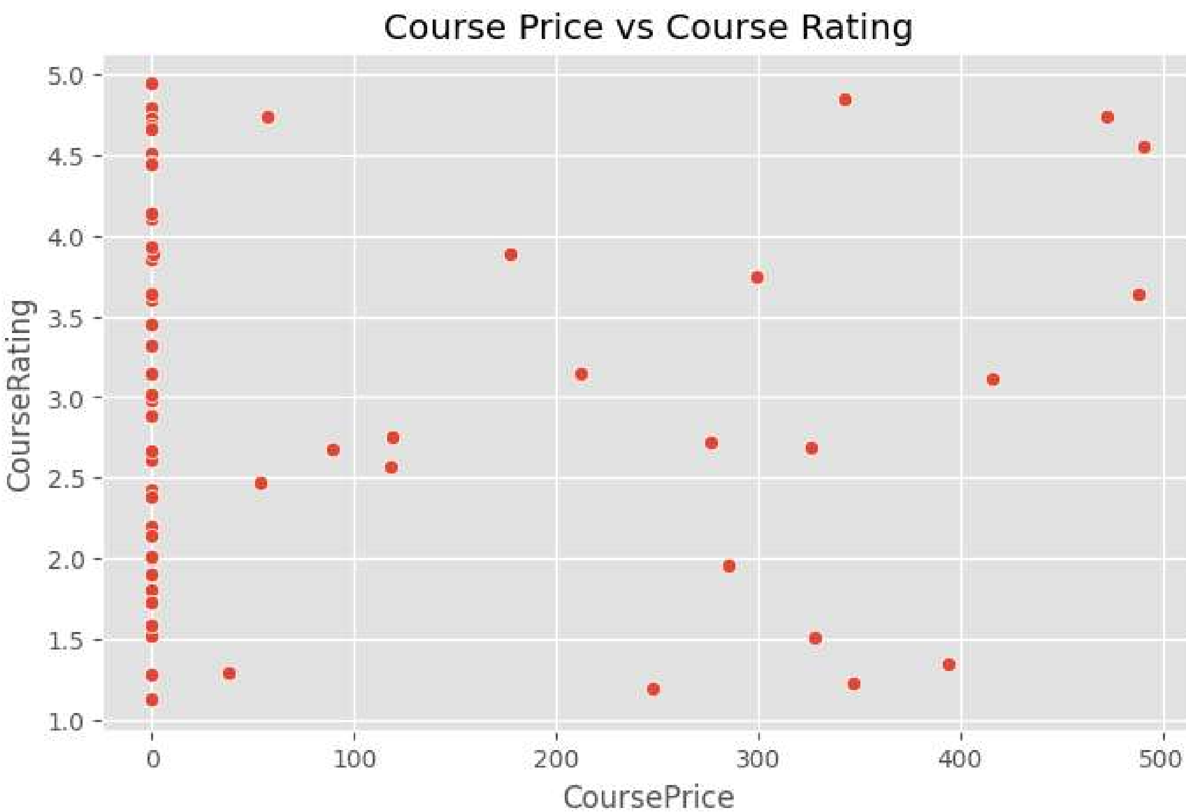
In [33]:
```
teacher_enrollment["RatingTier"] = pd.cut(
    teacher_enrollment["TeacherRating"],
    bins=[0,3.5,4.5,5],
    labels=["Low","Medium","High"])
```

```
tier_enrollment = (
    teacher_enrollment.groupby("RatingTier")
    ["EnrollmentCount"]
    .mean())
tier_enrollment.plot(kind="bar")
plt.title("Average Enrollment by Rating Tier")
plt.ylabel("Average Enrollment")
plt.show()
```



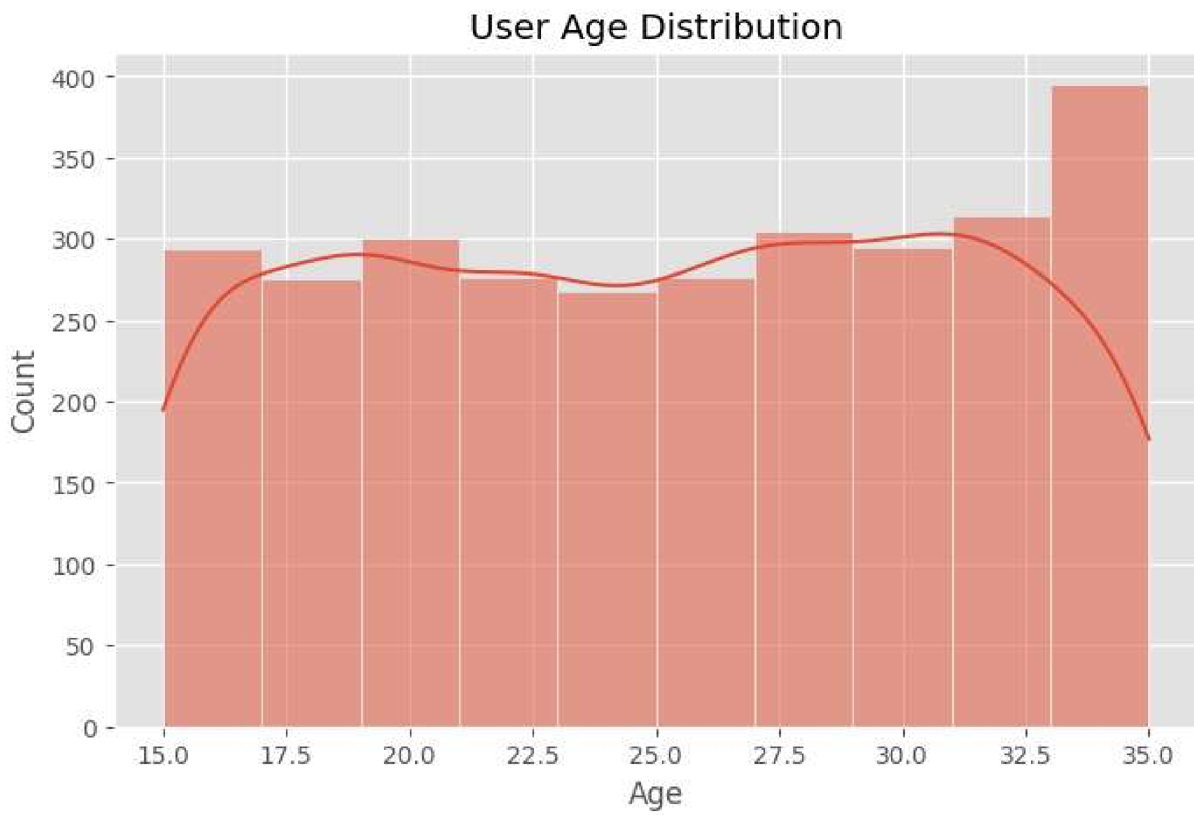
Course Price and Rating Relationship Analysis

In [34]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(
    data=courses,
    x="CoursePrice",
    y="CourseRating")
plt.title("Course Price vs Course Rating")
plt.show()
```



User Age Distribution Analysis

In [35]:
```
plt.figure(figsize=(8,5))
sns.histplot(
    users["Age"],
    bins=10,
    kde=True)
plt.title("User Age Distribution")
plt.show()
```



Key Findings

1. Teacher experience shows a positive relationship with instructor ratings.
2. Certain expertise domains consistently achieve higher learner ratings.
3. Intermediate level courses receive the highest average ratings.
4. Highly rated instructors tend to attract greater enrollments.
5. Course quality varies across categories and expertise areas.

Recommendations

1. Invest in instructor development programs.
2. Prioritize high-performing expertise domains.
3. Replicate teaching practices of top-rated instructors.
4. Monitor low-performing categories for quality improvement.
5. Use instructor ratings as a key quality benchmark.

Executive Insights

Key Business Observations

1. Teaching experience demonstrates a positive association with instructor ratings.

2. High-performing instructors contribute significantly to enrollment growth.

3. Certain expertise domains consistently achieve higher learner satisfaction scores.

4. Course quality varies across categories and learning levels.

5. Instructor quality serves as a leading indicator of course success.

Strategic Actions

- Develop mentorship programs for new instructors.
- Expand courses within high-performing expertise domains.
- Establish continuous instructor evaluation systems.
- Monitor low-performing categories for quality improvement.
- Reward top-rated instructors through recognition programs.

Research Summary

This study evaluated instructor performance and course quality using instructor demographics, expertise, experience, enrollment behavior, and course ratings.

Key outcomes indicate that instructor experience positively influences ratings, while highly rated instructors attract greater enrollment volumes. Expertise domains also demonstrate measurable differences in performance and learner satisfaction.

The findings support the implementation of a structured instructor evaluation framework for improving educational quality and learner outcomes.

Conclusion

The analysis demonstrates that instructor quality has a measurable impact on course performance and learner engagement. Experienced instructors generally achieve higher ratings, while high-rated instructors attract greater enrollment volumes. Expertise domains also show significant variation in performance. These findings support the adoption of a data-driven instructor evaluation framework for improving educational outcomes on the EduPro platform.